

Quadrilateral

Q1: The angle of the quadrilateral are in the ratio 3:5:9:13. Find all the angle of quadrilateral.

Sol: Let the angles be $3x^\circ$, $5x^\circ$, $9x^\circ$ and $13x^\circ$

Then $3x^\circ + 5x^\circ + 9x^\circ + 13x^\circ = 360^\circ$ [Sum of angles of Quadrilateral = 360°]

$$\Rightarrow 30x = 360^\circ$$

$$x = \frac{360}{30} = 12^\circ$$

$\therefore$ The angles are $3 \times 12 = 36^\circ$, $5 \times 12 = 60^\circ$
 $9 \times 12 = 108^\circ$, $13 \times 12 = 156^\circ$.

Q2: If the diagonals of parallelogram are equal, then show that it is a rectangle.

Sol: Given: A parallelogram ABCD in which $AC = BD$.

To prove: ABCD is a rectangle

Proof: In $\triangle ABC$ and $\triangle DCB$.

$$AB = CD \quad (\text{opp. side of } \parallel \text{gram})$$

$$BC = BC \quad (\text{Common})$$

$$AC = BD \quad (\text{Given})$$

$$\triangle ABC \cong \triangle DCB \quad [\text{By SSS}]$$

$$\Rightarrow \angle ABC = \angle DCB \quad [\text{By CPCT}]$$

$$\text{or } \angle B = \angle C$$

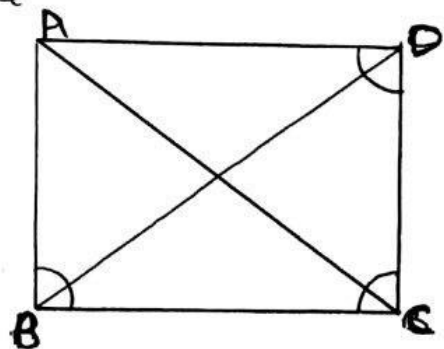
$$\text{Also } \angle B + \angle C = 180^\circ \quad [\text{co-interior angle}] [AB \parallel CD]$$

$$\Rightarrow \angle B + \angle B = 180$$

$$\Rightarrow 2\angle B = 180$$

$$\Rightarrow \angle B = \frac{180}{2} = 90^\circ$$

$$\angle B = \angle C = 90^\circ$$



ABCD is a ||gram. one of whose angle is 90°

Hence, ABCD is a rectangle.

Ques3: Show that if the diagonal of a quad. bisect each other at right angles, then it is a rhombus.

Sol: Given: A quad ABCD in which diagonal AC and BD intersect O such that

$$OA = OC, OB = OD \text{ and } AC \perp BD$$

To prove: ABCD is a rhombus.

Proof: In $\triangle AOB$ and $\triangle BOC$

$$AO = OC \text{ (Given)}$$

$$\angle 1 = \angle 2 \text{ (each } 90^\circ)$$

$$BO = BO \text{ (common)}$$

$$\therefore \triangle AOB \cong \triangle BOC \text{ (By SAS)}$$

$$AB = BC \text{ (By CPCT)}$$

Hence, ABCD is a rhombus.

[$\because$ If the diagonal of quad bisect each other, then it is a ||gram and opposite side of ||gram]. #

Ques4: Show that the diagonal of square are equal and bisect each other at right angles.

Sol: Given: ABCD is a square.

To prove: $AC = BD$

$$AC \perp BD$$

$$AO = OC, OB = OD.$$

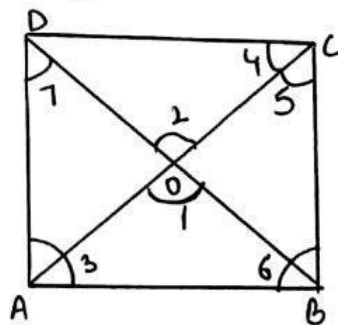
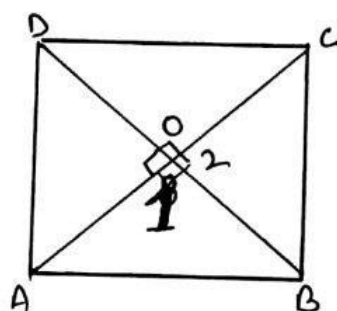
Proof: In $\triangle ABC$ and $\triangle BAD$.

$$AB = AB$$

$$BC = AD \text{ (side of square)}$$

$$\angle ABC = \angle BAD \text{ (each } 90^\circ)$$

$$\triangle ABC \cong \triangle BAD \text{ (By SAS)} \Rightarrow AC = BD \text{ (By CPCT)}$$



In $\triangle AOB$ and $\triangle COD$

$$AB = CD \quad (\text{Side of square})$$

$$\angle 1 = \angle 2 \quad (\text{V.O.A})$$

$$\angle 3 = \angle 4 \quad (\text{A.I.A})$$

$$\triangle AOB \cong \triangle COD. \quad (\text{By ASA})$$

$$\Rightarrow AO = OC \quad \# \quad \text{and} \quad OB = OD \quad \# \quad (\text{By CPCT})$$

In $\triangle ABC$

$$\angle 3 + \angle B + \angle 5 = 180^\circ$$

$$\Rightarrow \angle 3 + 90^\circ + \angle 3 = 180^\circ \quad \left[\begin{array}{l} AB = BC \\ \Rightarrow \angle 3 = \angle 5 \end{array} \right]$$

$$\Rightarrow 2\angle 3 = 180 - 90$$

$$\Rightarrow \angle 3 = \frac{90}{2} = 45^\circ$$

$$\Rightarrow \angle 3 = \angle 5 = 45^\circ$$

$$\text{Similarly } \angle 6 = \angle 7 = 45^\circ$$

In $\triangle AOB$

$$\angle 3 + \angle 1 + \angle 6 = 180^\circ$$

$$45^\circ + \angle 1 + 45^\circ = 180$$

$$\angle 1 = 180 - 45 - 45 = 90^\circ$$

$$\angle BOC = 90^\circ \Rightarrow BO \perp AC.$$

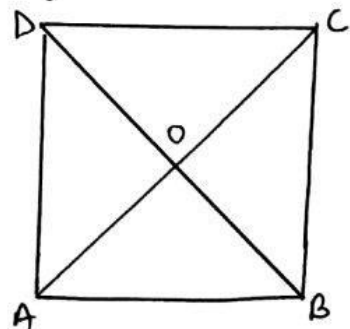
$$\Rightarrow BD \perp AC \quad \#$$

Ques 5: Show that if the diagonals of a quad. are equal and bisect each other at rt. angles, then it is a square

Given: A quad. $ABCD$, in which
 $AC = BD$ and $OA = OC$
 $BD \perp AC$ $OB = OD$

To prove: $ABCD$ is a square.

Proof: $ABCD$ is a quad. whose diagonals bisect each other, so it is a || gram.



Also, its diagonal bisect each other at right angle.

$\therefore$ ABCD is a rhombus.

$$\Rightarrow AB = BC = CD = DA.$$

In $\triangle ABC$ and $\triangle BAD$.

$$AB = AB \quad (\text{common})$$

$$BC = AD \quad (\text{Proved above})$$

$$AC = BD \quad (\text{Given})$$

$$\therefore \triangle ABC \cong \triangle BAD \quad (\text{By SSS})$$

$$\Rightarrow \angle ABC = \angle BAD \quad (\text{By CPCT})$$

$$\Rightarrow \angle A = \angle B = x \quad (\text{let})$$

$$\angle A + \angle B = 180^\circ$$

$$x + x = 180$$

$$2x = 180$$

$$x = \frac{180}{2} = 90^\circ$$

$$\angle A = \angle B = \angle C = \angle D = 90^\circ \quad [\text{opp. ~~sides~~ angle of ||gram}]$$

$\Rightarrow$ ABCD is a rhombus whose angle are of 90° each

Hence ABCD is a square.

Ques 6:- Diagonal AC of parallelogram ABCD. bisect $\angle A$

(i) it bisect $\angle C$

(ii) ABCD is a rhombus.

Given: In ||gram ABCD

$$\angle 1 = \angle 2.$$

Proof: $AB \parallel CD$ and $BC \parallel AD$

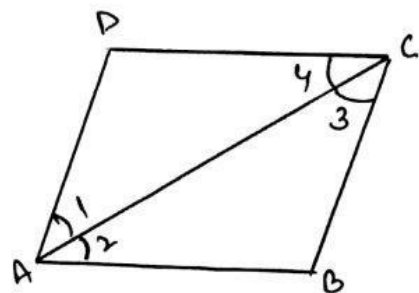
$$\Rightarrow \angle 1 = \angle 3 - \text{①} \quad [\text{A.I.A}]$$

$$\angle 2 = \angle 4 - \text{②}$$

$$\text{Given } \angle 1 = \angle 2 - \text{③}$$

$$\text{From ① ② ③ } \Rightarrow \angle 3 = \angle 4.$$

$\therefore$ AC bisect $\angle C$.



In $\triangle ABC$ and $\triangle ADC$ From ① ② ③

$$AC = AC$$

$$\angle 1 = \angle 2 = \angle 3 = \angle 4$$

In $\triangle ABC$, $\angle 2 = \angle 3$

$$\Rightarrow AB = BC$$

In $\triangle ADC$, $\angle 1 = \angle 4$

$$\Rightarrow AD = CD$$

Also, $ABCD$ is a $\parallel\text{gm}$.

$$AB = CD, AD = BC.$$

$$\therefore AB = BC = CD = DA$$

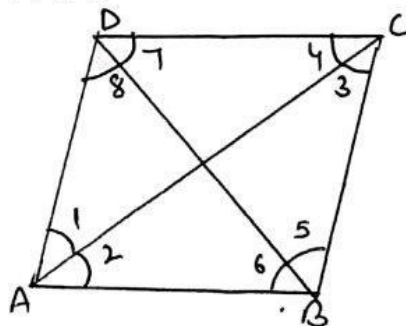
Hence, $ABCD$ is a rhombus.

Q7 $ABCD$ is a rhombus. Show that the diagonal AC bisect $\angle A$ as well as $\angle C$ and diagonal BD bisect $\angle B$ as well as $\angle D$

Sol: Given: $ABCD$ is a rhombus. i.e.

$$AB = BC = CD = DA$$

To prove: $\angle 1 = \angle 2$
 $\angle 3 = \angle 4$
 $\angle 5 = \angle 6$
 $\angle 7 = \angle 8$



In $\triangle ABC$ and CDA

$$AB = AD \text{ (Given)}$$

$$AC = AC \text{ (common)}$$

$$BC = DA \text{ (Given)}$$

$$\triangle ABC \cong \triangle CDA \quad [\text{By SSS}]$$

$$\Rightarrow \angle 1 = \angle 2 \text{ and } \angle 3 = \angle 4 \quad [\text{By CPCT}]$$

Similarly $\angle 5 = \angle 6$ and $\angle 7 = \angle 8$.

Hence diagonal AC bisect $\angle A$ and $\angle C$
and diagonal BD bisect $\angle B$ and $\angle D$.

Q8: ABCD is a rectangle in which diagonal AC bisect $\angle A$ as well as $\angle C$. Show that (i) ABCD is a square.
 (ii) Diagonal BD bisect $\angle B$ as well as $\angle D$.

Sol: Given: ABCD is a rectangle.

AC bisect $\angle A$ and $\angle C$.

To prove: (i) ABCD is a square.

(ii) Diagonal BD bisect $\angle B$ as well as $\angle D$.

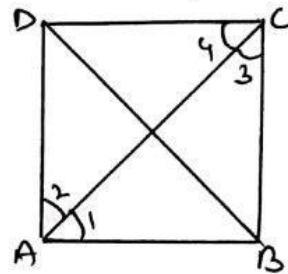
Proof: (i) Since AC bisect $\angle A$ and $\angle C$.

$$\angle 1 = \angle 2 = \angle 3 = \angle 4 \quad [\because \text{each } \frac{90}{2} = 45^\circ]$$

In $\triangle ABC$

$$\angle 1 = \angle 3$$

$\Rightarrow AB = BC$. (side opp. to equal angle)



In rectangle opp. side are equal

$$\therefore AB = BC = CD = DA$$

Thus, ABCD is a square.

(ii) In square, diagonal bisect the angle D and B.

So, BD bisect $\angle B$ as well as $\angle D$.

Q9: In a parallelogram ABCD, two point P and Q are taken on diagonal BD such that $DP = BQ$. Show that

$$(i) \triangle APD \cong \triangle CQB$$

$$(ii) AP = CQ$$

$$(iii) \triangle AQB = \triangle CPD$$

$$(iv) AQ = CP$$

(v) APCQ is a parallelogram.

Given : ABCD is a ||gram

P and Q are point on
diagonal BD

Such that $DP = BQ$

To prove : _____

Proof :

(i) In $\triangle APD$ and $\triangle CQB$.

$$AD = BC \quad (\text{opp. side of ||gram})$$

$$DP = BQ \quad (\text{Given})$$

$$\angle 1 = \angle 2 \quad (\text{A.I.A})$$

$$\triangle APD \cong \triangle CQB \quad (\text{By SAS})$$

$$(ii) \therefore AP = CQ \quad [\text{CPCT}]$$

(iii) In $\triangle AQB$ and $\triangle CPD$, we have.

$$AP = CQ \quad (\text{opp. side of ||gram})$$

$$DP = BQ \quad (\text{Given})$$

$$\angle 3 = \angle 4 \quad (\text{A.I.A})$$

$$\triangle AQB \cong \triangle CPD \quad (\text{By SAS})$$

$$(iv) \therefore AQ = CP \quad [\text{By CPCT}]$$

(v) Since in $APCQ$, opposite side are equal
 $\therefore$ it is a ||gram.

Q10. ABCD is a ||gram and AP and CQ are line from
vertices A and C on diagonal BD respectively.

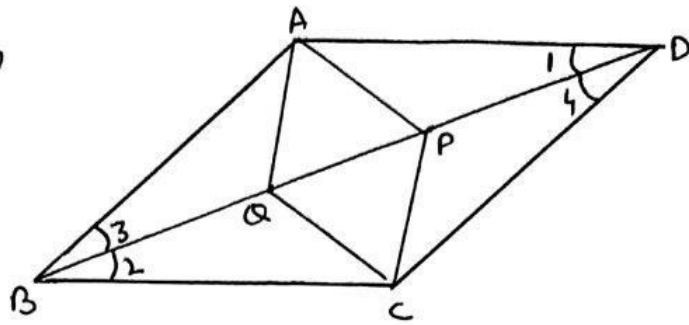
Show that (i) $\triangle APB \cong \triangle CQD$.

(ii) $AP = CQ$.

Sol.:

Given ABCD is a ||gram

AP and CQ are line from vertices A and C.
on diagonal BD.



Proof: In $\triangle APB$ and $\triangle CQD$

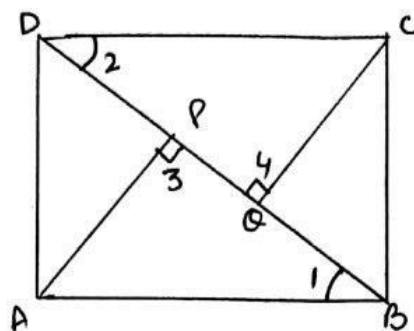
$$\angle 1 = \angle 2 \text{ (A.I.A)}$$

$$AB = CD \text{ (opp. side of ||gram)}$$

$$\angle 3 = \angle 4 \text{ (each } 90^\circ)$$

$$\triangle APB \cong \triangle CQD \text{ (ASA).}$$

$$\text{So } AP = CQ. \text{ [Proved]} \\ \text{[CPCT]}$$



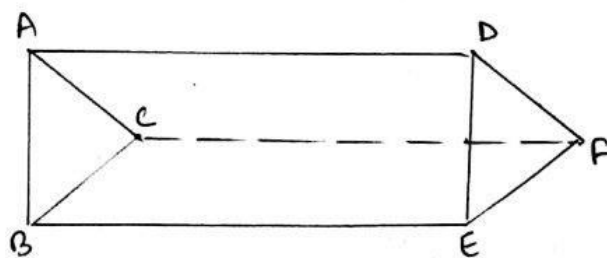
Q11. In $\triangle ABC$ and $\triangle DEF$, $AB = DE$, $AB \parallel DE$, $BC = EF$ and $BC \parallel EF$. Vertices A, B, C are joined to vertices D, E, F. Show that

- (1) Quad ABED is a ||gram
- (2) Quad BEFC is a ||gram
- (3) $AD \parallel CF$ and $AD = CF$.
- (4) Quad ACFD is a ||gram
- (5) $AC = DF$
- (6) $\triangle ABC \cong \triangle DEF$.

Sol :- Given : Two $\triangle ABC$ and $\triangle DEF$ such that

$$AB = DE \text{ and } AB \parallel DE$$

$$BC = EF \text{ and } BC \parallel EF$$



To prove: See as above

Proof: 1) In quad ABED, $AB = DE$ and $AB \parallel DE$
one pair of opp. side is equal and ||
 $\Rightarrow$ ABED is a ||gram.

(ii) In quad $BEFC$, we have $BC = EF$ and $BC \parallel EF$
 one pair of opp. side is equal and $\parallel$.
 $\therefore BEFC$ is a $\parallel$ gram.

Now (iii) $AD = BE$ and $AD \parallel BE$ — (1)
 $[\because ABED \text{ is a } \parallel\text{gram}]$

and $CF = BE$ and $CF \parallel BE$ — (2)
 $[\because BEFC \text{ is a } \parallel\text{gram}]$

From (1) and (2) $AD = CF$ and $AD \parallel CF$.

$\Rightarrow$ one pair of opp. side is equal and $\parallel$

$\Rightarrow ACFD$ is a $\parallel$ gram. [Proved iv]

(v) Since $ACFD$ is a parallelogram.

$AC = DF$ [opp. side of $\parallel$ gm $ACFD$]

(vi) In $\triangle ABC$ and $\triangle DEF$, we have.

$AB = DE$ (opp. side of $\parallel$ gram)

$BC = EF$ "

$CA = FD$ "

$\therefore \triangle ABC \cong \triangle DEF$ (By SSS)

Q12:- $ABCD$ is a trapezium in which

$AB \parallel CD$ and $AD = BC$. Show that

(i) $\angle A = \angle B$

(ii) $\angle C = \angle D$

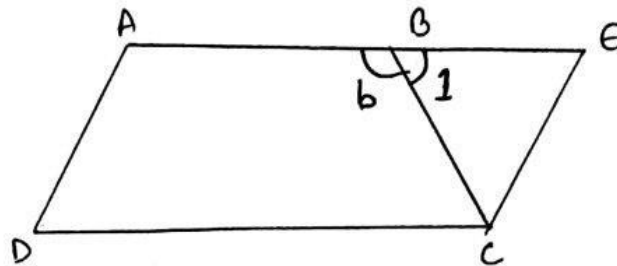
(iii) $\triangle ABC = \triangle BAD$.

(iv) Diagonal $AC =$ Diagonal BD .

Given : ABCD is a trapezium in which $AB \parallel CD$
and $AD = BC$

To prove: See as above . Const:- Produce AB and draw a line $CE \parallel AD$

Proof :



Since $AB \parallel DC$ -

$\Rightarrow AE \parallel DC$ - (i)

and $AD \parallel CE$ - (ii)

$\Rightarrow ADCE$ is a ||gram. [opp. side of ||gram are ||el]

$\angle A + \angle E = 180$ - (iii) [consecutive interior angle]

$\angle B + \angle 1 = 180$ - (iv) [linear pair]

$AD = CE$ [opp. side of ||gram are equal] - (5)

$BC = CE$ - (6) [Given]

From (5) and (6)

$AD = BC$

$\Rightarrow \angle 1 = \angle E$ - (7) [angle opp. to equal side are equal]

From (iii) and (7)

$\angle A + \angle 1 = 180$ - (8)

From (iv) and (8)

$\angle B + \angle 1 = \angle A + \angle 1$

$\angle B = \angle A$ [Proved.]

(ii) $\angle A + \angle D = 180^\circ$

$\angle B + \angle C = 180^\circ$

$\Rightarrow \angle A + \angle D = \angle B + \angle C$

$\Rightarrow \angle D = \angle C$ [$\because \angle A = \angle B$]

(iii) In $\triangle ABC$ and $\triangle BAD$

$AD = BC$ (Given)

$\angle A = \angle B$ [Proved]

$$AB = AB \text{ (common)}$$

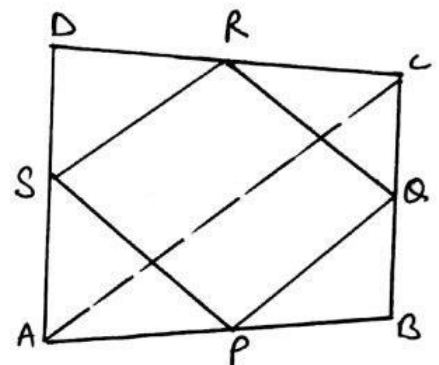
$$\therefore \triangle ABC \cong \triangle BAD \text{ [By ASA]}$$

$$(iv) \text{ diagonal } AC = \text{diagonal } BD \text{ [By CPCT].}$$

Exercise: 8.2

Q1. ABCD is a quadrilateral in which P, Q, R and S are mid point of side AB, BC, CD and DA respectively. AC is a diagonal. Show that

- (i) $SR \parallel AC$ and $SR = \frac{1}{2} AC$.
- (ii) $PQ = SR$
- (iii) PQRS is a parallelogram.



Given: See as above.

To prove: See as above

Proof: In $\triangle ABC$, P is the mid point of AB and Q is the mid point of BC.

$$\therefore PQ \parallel AC \text{ and } PQ = \frac{1}{2} AC \text{ — (1) [By Midpoint Theorem]}$$

In $\triangle ADC$, R is the mid point of CD and S is the mid point of AD.

$$\therefore SR \parallel AC \text{ and } SR = \frac{1}{2} AC \text{ — (2) [Proved (1)]}$$

[By Mid point theorem]

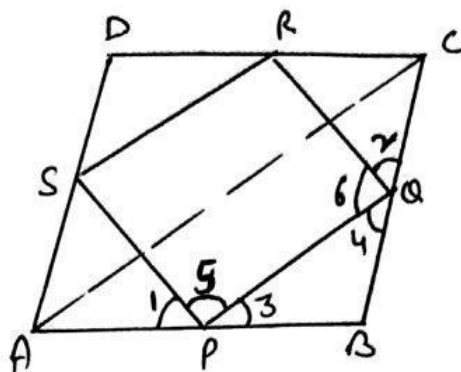
(i) From (1) and (2)

$$PQ \parallel SR \text{ and } PQ = SR$$

(ii) Now in quadrilateral PQRS, its one pair of opposite side is equal and $\parallel$ el.

$\therefore$ PQRS is a parallelogram.

Q2. ABCD is a rhombus and P, Q, R, S are the mid point of the sides AB, BC, CD and DA respectively. Show that the quad. PQRS is a rectangle.



Sol: Given :-

To prove : PQRS is a rectangle.

Const. Join AC.

Proof:- In $\triangle ABC$, P and Q are mid pt. of side AB and BC.

$$\therefore PQ \parallel AC \text{ and } PQ = \frac{1}{2} AC \text{ --- (1)}$$

In $\triangle ADC$, R and S are mid point of side CD and AD.

$$\therefore SR \parallel AC \text{ and } SR = \frac{1}{2} AC \text{ --- (2)}$$

From (1) and (ii)

$$PQ \parallel RS \text{ and } PQ = SR$$

In quad PQRS, one pair of opposite side is equal and parallel.

$\therefore$ PQRS is a parallelogram.

$$\therefore AB = BC$$

$$\frac{1}{2} AB = \frac{1}{2} BC$$

$$\Rightarrow PQ = QR$$

$$\Rightarrow \angle 3 = \angle 4 \text{ --- (3)}$$

[angle opp. to equal side are equal] .

In $\triangle APS$ and $\triangle CQR$.

$$\begin{aligned} AP &= CQ \\ AS &= CR \\ PS &= QR \end{aligned} \quad \begin{aligned} &[\text{Half of equal side } AB, BC \\ &\text{and } AD, CD] \\ &[\text{Opp. side of } \parallel \text{gram}] \end{aligned}$$

$$\triangle APS \cong \triangle CQR \quad [\text{SSS}]$$
$$\angle 1 = \angle 2 \quad (\text{By CPCT}) \quad \text{--- (1)}$$

$$\angle 1 + \angle 5 + \angle 3 = 180^\circ \quad \text{--- (2)}$$

$$\angle 2 + \angle 4 + \angle 6 = 180^\circ \quad \text{--- (3)}$$

$$\angle 1 + \angle 5 + \angle 3 = \angle 2 + \angle 4 + \angle 6$$

$$\Rightarrow \angle 5 = \angle 6 \quad \left[\because \begin{array}{l} \angle 1 = \angle 2 \\ \angle 3 = \angle 4 \end{array} \right]$$

$$(\text{or}) \Rightarrow \angle SPQ = \angle PQR$$

Also $SP \parallel RQ$.

$$\angle SPQ + \angle PQR = 180^\circ$$

$$\angle SPQ + \angle SPQ = 180$$

$$2\angle SPQ = 180$$

$$\angle SPQ = \frac{180}{2} = 90^\circ$$

$$\Rightarrow \angle SPQ = \angle PQR = 90^\circ$$

Thus, $PQRS$ is a rectangle where one angle $\angle SPQ = 90^\circ$ #

Q3:- $ABCD$ is a rectangle. and P, Q, R and S are mid point of the side AB, BC, CD, DA . Show that quadrilateral $PQRS$ is a rhombus.

Sol:-
Given:-

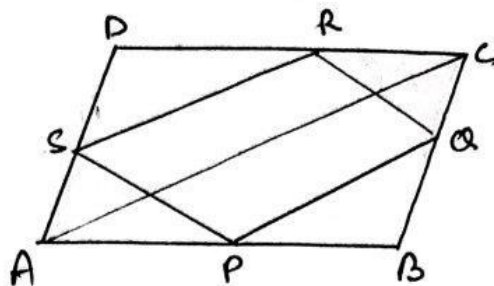
To prove: $PQRS$ is a rhombus

Const: Join AC .

Given:-

To Prove: In $\triangle ABC$,

P and Q are mid-point
of side AB and BC.



$$PQ \parallel AC, PQ = \frac{1}{2} AC \quad \text{--- (1)}$$

Similarly in $\triangle ADC$, R and S are mid point of
side CD and AD.

$$\therefore SR \parallel AC \text{ and } SR = \frac{1}{2} AC. \quad \text{--- (2)}$$

From (2) and (1)

$$PQ \parallel SR \text{ and } PQ = SR. \quad \text{--- (3)}$$

Now in quad. PQRS, one pair of opposite
side PQ and SR is parallel and equal.

$\therefore$ PQRS is a parallelogram --- (4)

$$\text{Now } AD = BC$$

$$\Rightarrow \frac{1}{2} AD = \frac{1}{2} BC$$

$$\Rightarrow AS = BQ,$$

In $\triangle APS$ and $\triangle BQP$

$$\angle A = \angle B \quad [\text{each } 90^\circ]$$

$$AS = BQ$$

$$AP = BP \quad [P \text{ is the mid point}]$$

$$\triangle APS \cong \triangle BQP$$

$$\Rightarrow PS = PQ \quad [\text{CPCT}] \quad \text{--- (5)}$$

From (4) and (5), we get

$$PS = PQ = QR = SR.$$

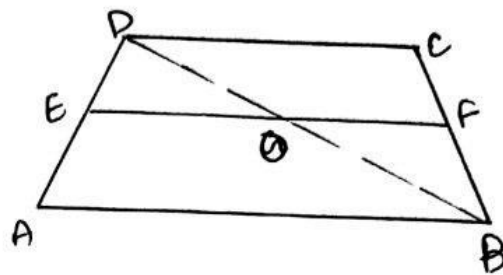
$\Rightarrow$ PQRS is a rhombus.

Q4: ABCD is a trapezium in which $AB \parallel DC$. A line is drawn through E parallel to AB intersecting BC at F. Show that F is the mid point of BC.

Sol: In $\triangle DAB$,

E is the mid point of AD.

$ED \parallel AB$ [$\because EF \parallel AB$]



$\therefore$ By converse of mid pt. theorem, O is the mid point of BD.

In $\triangle BCD$, O is the mid point of BD.

$OF \parallel CD$.

$\therefore$ By converse of mid point theorem.

F is the mid point BC.

Ques 5: In a parallelogram ABCD, E and F are mid point of side AB and CD. Show that the line segment AF and EC intersect diagonal BD.

Sol: To prove: $BQ = QP = PD$.

Proof: Since E and F are mid point of AB and CD.

$$\Rightarrow AE = \frac{1}{2} AB$$

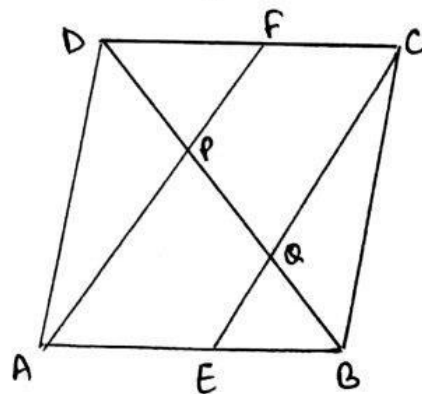
$$\text{and } CF = \frac{1}{2} DC$$

But $AB = CD$ and $AB \parallel CD$.

$\therefore AE = CF$ and $AE \parallel CF$.

$\Rightarrow AECF$ is a parallelogram.

[One pair of opp. side is $\parallel$ and equal]



In $\triangle BAP$.

E is the mid point of AB

$$EQ \parallel AP.$$

$\Rightarrow Q$ is the mid point of PB

$$\Rightarrow PQ = QB. \quad - (1)$$

Similarly, in $\triangle DQC$.

P is the mid point of DQ

$$DP = PQ. \quad - (2)$$

From (1) and (2)

$$DP = PQ = QB \quad \#$$

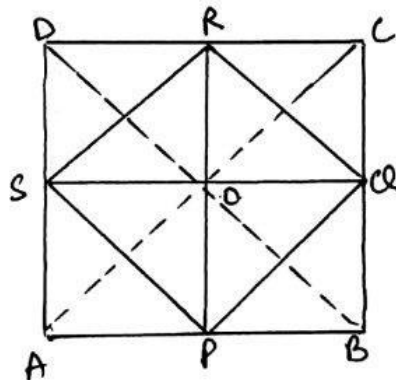
~~Q~~ ~~line~~.

Ques 6. Show that the line segment joining the mid point of opposite side of a quadrilateral bisect each other.

Sol: Const. Join PQ, QR, RS, SP, AC, BD.

Proof: In $\triangle ABC$, P and Q are mid point of AB and BC.

$$\therefore PQ \parallel AC \text{ and } PQ = \frac{1}{2} AC$$



In $\triangle ACD$, S and R are mid point of AD and CD.

$$RS \parallel AC \text{ and } RS = \frac{1}{2} AC$$

$$\therefore PQ \parallel SR \text{ and } PQ = SR.$$

Thus, a pair of opposite side of 'a quad. PQRS are parallel and equal.

$\therefore PQRS$ is a parallelogram

And in parallelogram, diagonal bisect each other

$$\Rightarrow OP = OR, OQ = OS.$$

Ques 7. ABC is a triangle right angle at C . A line through the mid point M of hypotenuse AB and parallel to BC intersect AC at D . Show that:

- (i) D is the mid point AC .
- (ii) $MD \perp AC$.
- (iii) $CM = MA = \frac{1}{2} AB$.

Proof:- In $\triangle ABC$, M is the mid point of AB and $MD \parallel BC$.

$\therefore D$ is the mid point of AC .

$$\Rightarrow AD = DC.$$

(ii) Since $MD \parallel BC$

$$\angle ADM = \angle ACB = 90^\circ \text{ [Corresponding angles]}$$

$$\Rightarrow MD \perp AC.$$

(iii) In $\triangle AMD$ and $\triangle CMD$,

$$AD = CD$$

$$\angle ADM = \angle CDM$$

$$MD = MD \text{ (Common)}$$

$$\triangle AMD \cong \triangle CMD.$$

$$\Rightarrow MA = MC. \text{ [By CPCT]}$$

$$MA = \frac{1}{2} AB. \text{ Since } M \text{ is the mid point of } AC.$$

$$\text{Hence } CM = MA = \frac{1}{2} AB. \quad \#$$

